

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Contributed to the **review of system requirements and architecture design** for automotive camera systems, assessing testability and planning coverage in collaboration with System Engineers.
- Designed **test specifications** to fulfill system functional and performance requirements, ensuring comprehensive coverage against system level standards and requirements.
- Implemented **test automation** capabilities following platform standards and project automation targets, executing tests in bench and vehicle environments.
- Analyzed test results, test logs, and reports, maintaining traceability of system test specifications to system requirements throughout the project lifecycle.
- Participated in **cross-functional team** collaboration for requirements gathering and UAT support, contributing to presentations and maintaining professional working relationships with internal departments.
- Developed **Python scripts** for automation and data pipelines, supporting scheduled reporting and ETL processes within the HCL Technologies enterprise systems support team.
- Containerized applications using **Docker** and deployed to **AWS EC2** instances, managing CI/CD pipelines with **Jenkins** and Git for code reviews and version control.

SKILLS

Programming Languages – Python, C++, C, C#, .NET, NI LabVIEW, NI TestStand

Automotive Systems – ADAS, camera systems, mechatronic ECU systems, automotive software features

Testing & Validation – System validation, functional requirements, performance requirements, test environment specification, test methods, system level testing, stress testing, robustness testing, trace analysis, root cause analysis

Cloud & DevOps – AWS EC2, Docker, Jenkins, Git

Methodologies – Systems Development Life Cycle, Agile, Scrum

PROJECTS

Automotive Camera System Test Automation Framework

- Developed an automated test framework using **Python** and **pytest** to execute system-level tests for automotive camera features, covering functional and performance requirements.
- Integrated the framework with **Jenkins CI/CD** pipelines to enable continuous integration and automated regression testing, reducing manual test execution time by **40%**.
- Contributed to the design of test environment specifications for bench testing, ensuring readiness for planned system development phases.

Embedded Software Validation Suite

- Designed and implemented test specifications for mechatronic ECU systems, focusing on validating software features against system requirements and standards.
- Executed system testing aligned with project schedules, analyzing test logs and reporting defects for assigned features to ensure test completion status.
- Maintained test results and reports throughout the project lifecycle, ensuring traceability from system requirements to test case execution.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)